1. Structured Spaces

This semester, you will be introduced to certain structures that can be given to a simple set (which we will often call a *„space“* X in this context). You can think of this as a hierarchy, where each level grants us more powerful mathematical tools:

a) **Topological spaces:** (Focus in the 4th semester)

b) **Metric spaces** (X, d)**:** Defines a mere *distance* between points. Important: X does not necessarily have to be a vector space here! (Focus in Analysis)

c) **Normed vector spaces** $(X, \|\cdot\|)$**:** Defines the *length* of a vector. This presupposes a linear structure (i.e., X must be a vector space). (Focus in LinAlg / Analysis)

d) **Inner product spaces** $(X, \langle\cdot,\cdot\rangle)$**:** Defines both lengths and *angles*! This is the richest structure. (Focus in LinAlg)

[Gemini-Notiz: Why is this hierarchy important? Because the more powerful structures automatically induce the weaker ones! An inner product always induces a norm via the formula $\|x\| := \sqrt{\langle x, x\rangle}$. And a norm always induces a metric via the formula $d(x, y) := \|x - y\|$. The reverse, however, is not generally true!]

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2. Metric Spaces

A **metric space** (X, d) is any set X equipped with a **metric**. You can think of it as a universal ruler.

Formally, a metric is a function:

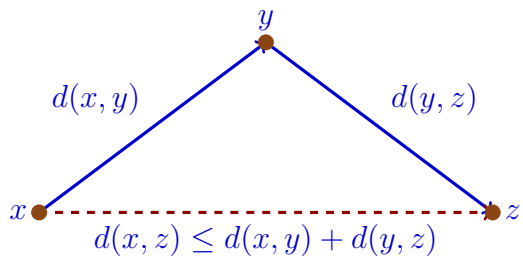
$$d : X \times X \to \mathbb{R}_{\geq 0}$$

such that for all points $x, y, z \in X$, the following three intuitive properties hold:

a) **Definiteness:** $d(x, y) \geq 0$, and $d(x, y) = 0 \iff x = y$.
In words: *„The distance is always positive, and a point has zero distance only to itself.“*

b) **Symmetry:** $d(x, y) = d(y, x)$.
In words: *„The path from you to me is exactly as long as the path from me to you.“*

c) **Triangle Inequality:** $d(x, z) \leq d(x, y) + d(y, z)$.
In words: *„Taking a detour through a third point can never shorten your trip.“*



Examples of Metric Spaces

Example (The Discrete Metric)**.** Let X be any set. The discrete metric is the *„digital“* approach to measuring distance:

$$d(x,y) = \begin{cases} 1 & \text{if } x \neq y \\ 0 & \text{if } x = y \end{cases}$$

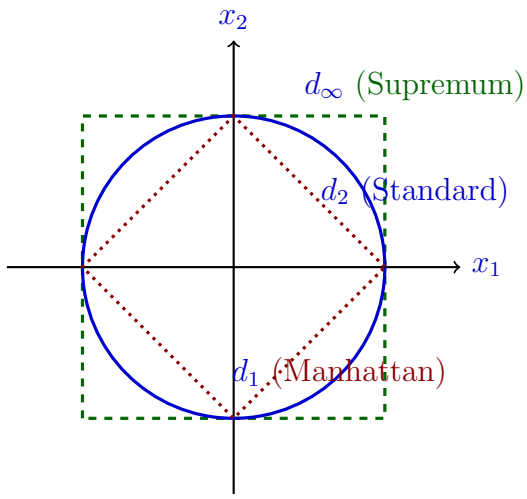
[Gemini-Notiz: This metric is an extremely important tool for constructing counterexamples! It perfectly demonstrates that a metric does not have to match our intuitive geometric understanding of space.]

Example (Metrics in $\mathbb{R}^n$)**.** We have many choices for measuring distance in $\mathbb{R}^n$. Some common ones are:

- **Manhattan metric (d_1):** $d_1(x,y) = \sum_{k=1}^{n} |x_k - y_k|$ (Like a taxi navigating a city grid)
- **Standard/Euclidean metric (d_2):** $d_2(x,y) = \sqrt{\sum_{k=1}^{n} (x_k - y_k)^2}$ (The classical straight-line distance)
- *„Supremum metric“* **(d_3 or d_∞):** $d_3(x,y) = \sup_{k=1,\ldots,n} |x_k - y_k|$ (Only the single largest coordinate difference matters)

As a generalization: $d_m(x,y) = \sqrt[m]{\sum_{k=1}^{n} |x_k - y_k|^m}$ is a valid metric on $\mathbb{R}^n$ for any $m \geq 1$.

Visualizing the *„Unit Balls“* in $\mathbb{R}^2$:

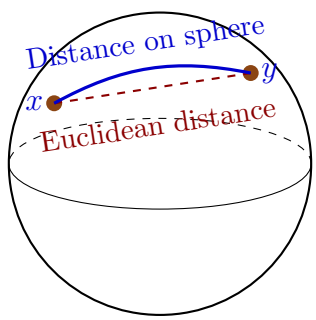


Example (Continuous Functions)**.** Let $X = C^0([0,1], \mathbb{R})$ denote the set of all continuous functions mapping the interval $[0,1]$ to $\mathbb{R}$. We can equip this function space with the *„Supremum Metric“*:

$$d(f,g) := \sup_{x \in [0,1]} |f(x) - g(x)|$$

Intuition: This metric finds the single point x where the two graphs are furthest apart, and defines that maximum vertical gap as the distance between the two functions.

Example (The n-Sphere)**.** Let $X = S^2 := \{(x_1, x_2, x_3) \in \mathbb{R}^3 : x_1^2 + x_2^2 + x_3^2 = 1\}$ (the 2-dimensional sphere). The closest path between two points on the sphere is an arc of a great circle (a geodesic). This defines a unique metric on the surface, similar to how straight lines define a metric on $\mathbb{R}^n$.



[Gemini-Warnung: Terminology check: A sphere embedded in 3D space is actually a 2-dimensional surface, which is why topologists denote it as S^2. The script loosely refers to it as the *„n-sphere“* here based on the ambient space.]

Exercise. Show that the supremum metric from the continuous functions example is indeed a metric.

Solution. Symmetry and definiteness follow directly from the properties of the absolute value $|\cdot|$. For the triangle inequality, let's consider three functions $f, g, h \in C^0([0,1])$ and define two helpful auxiliary functions:

$$p := f - h, \quad q := h - g$$

For any arbitrary but fixed $x \in [0,1]$, the value of the function at that specific point can never exceed the supremum of the entire function:

$$|p(x)| \leq \sup|p| \quad \text{and} \quad |q(x)| \leq \sup|q|$$

If we now look at the distance between f and g at the point x, we can use the ädd zerottrick and apply our inequalities:

$$\begin{aligned} |f(x) - g(x)| &= |p(x) + q(x)| \\ &\leq |p(x)| + |q(x)| \quad \text{(Standard triangle inequality in } \mathbb{R}\text{)} \\ &\leq \sup|p| + \sup|q| \\ &= \sup|f - h| + \sup|h - g| \end{aligned} \tag{1}$$

Because this upper bound holds for every single $x \in [0,1]$, it must inherently also hold for the supremum taken over all x! Thus:

$$d(f,g) \leq d(f,h) + d(h,g)$$

This proves the triangle inequality. $\square$

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3. Open and Closed Sets

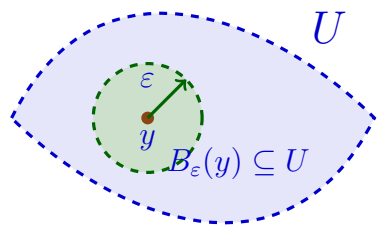
Let (X, d) be a metric space, let $x \in X$, and let $r \in \mathbb{R}_{>0}$. First, we define the *„open ball“* of radius r:

$$B_r(x) := \{y \in X : d(y,x) < r\}$$

Note: Pay attention to the strict inequality ($<$). The boundary shell is not part of the ball!

Definition.

- A subset $U \subseteq X$ is **open** iff for *every* $y \in U$, we can find an $\varepsilon > 0$ such that the entire ball $B_\varepsilon(y)$ lies completely inside U. (You can never accidentally step on the boundary.)
- A subset $A \subseteq X$ is **closed** iff its complement $X \setminus A$ is open.



Exercise. Let (X, d) be a metric space. Show that the entire space X and the empty set $\emptyset$ are both open and closed.

Solution. First, let's look at openness:

- X is open because for any $x \in X$ and any radius $r > 0$, the ball $B_r(x)$ is by definition a subset of X. Thus, $B_r(x) \subseteq X$ trivially holds.
- $\emptyset$ contains no points. Therefore, the condition *„for every point $y \in \emptyset$ there exists...“* is a vacuously true statement. Thus, the empty set is open.

Now for closedness: Since X and $\emptyset$ are each other's complements in X (i.e., $X \setminus X = \emptyset$ and $X \setminus \emptyset = X$), and we just proved both are open, they must both inherently be closed as well! $\square$

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4. Sequences and Convergence

A sequence $(x_n)_{n=0}^{\infty}$ in a metric space (X, d) is formally a function from $\mathbb{N} \to X$. We say that (x_n) **converges** to a limit $x \in X$ if:

$$\forall \varepsilon > 0, \exists N \in \mathbb{N} \text{ such that } d(x_n, x) < \varepsilon \quad \forall n \geq N$$

The Big Connection Proposition

Topological properties (openness/closedness) and the behavior of sequences go hand-in-hand:

a) $U \subseteq X$ is **open** $\iff$ for every sequence in X that has a limit in U, the sequence eventually lies entirely inside U.

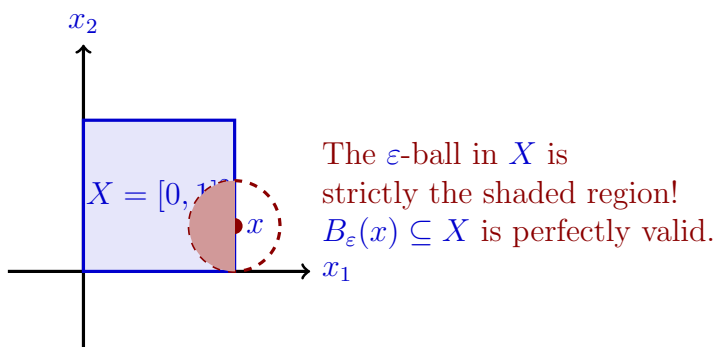
b) $A \subseteq X$ is **closed** $\iff$ *every* convergent sequence that lies entirely in A must necessarily have its limit in A as well. (The sequence cannot *„break out“* of the set.)

Watch out! A Common Misconception!

Let's look at the standard Euclidean metric space $\mathbb{R}^2$. We know the square $[0,1]^2$ is closed and $(0,1)^2$ is open. (The rectangle $[0,1] \times (0,1)$ is neither open nor closed in $\mathbb{R}^2$).

But... what if our *entire universe* (our metric space X) is defined strictly as $X := [0,1]^2$? In the subspace $X = [0,1]^2$, the set $[0,1]^2$ is suddenly *open*!

Why? Because the full space X is always open within itself. If you take a point exactly on the boundary, its open ε-ball consists only of points that actually exist in X. The ball does not bleed out into $\mathbb{R}^2$ because $\mathbb{R}^2$ simply does not exist for this specific metric space!



Exercise. **a)** Write down the definition of convergence for a sequence of functions $(f_n)_{n=0}^{\infty}$ to f in the metric space X of bounded functions $[0,1] \to \mathbb{R}$ equipped with the supremum metric.

b) You know this resulting expression from Analysis 1. What kind of convergence is this called?

c) Use the fact that the space of continuous functions $C^0([0,1]) \subseteq X$ is closed, and combine it with the *„sequentially closed“* proposition (part b) to conclude a massive result from Analysis 1.

Solution. **a)** $\forall \varepsilon > 0 \, \exists N \in \mathbb{N}$ such that $\sup_{x\in[0,1]} |f_n(x) - f(x)| < \varepsilon \quad \forall n \geq N$.

b) This is exactly the definition of **Uniform Convergence**!

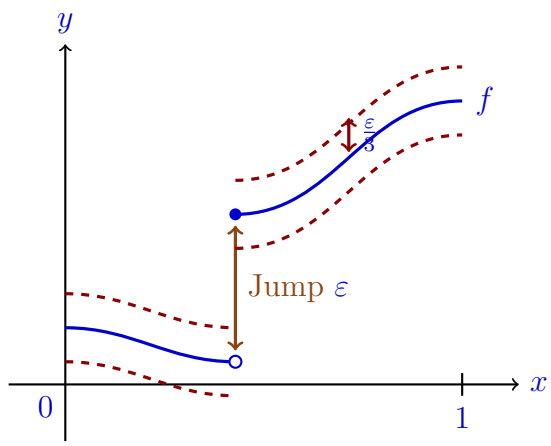
c) The fact that $C^0([0,1])$ is closed in the space of bounded functions means (according to our proposition above): If a sequence of continuous functions converges uniformly to a limit function f, then this limit function f cannot leave the space of continuous functions. Ergo: **The uniform limit of continuous functions is always continuous!**

Proof of closedness (The $\varepsilon/3$-trick): We show that the complement is open. Let $f \in X \setminus C^0([0,1])$. This means f is bounded but has at least one discontinuity at a point x_0. That implies there is a real jump $\varepsilon > 0$, such that no matter how close (δ) we get to x_0, we can always find a y where $|f(x_0) - f(y)| \geq \varepsilon$.

Now we consider an open ball around this discontinuous function f, but we give it a

tiny radius of $\varepsilon/3$:

$$B_{\varepsilon/3}(f) = \left\{ g \in X : \sup_{x \in [0,1]} |f(x) - g(x)| < \frac{\varepsilon}{3} \right\}$$



This ball (the „$\varepsilon/3$-*tube*“ around f) contains exclusively functions that are forced to replicate this jump. No function g trapped inside this tube can possibly be continuous! Therefore, the entire ball $B_{\varepsilon/3}(f)$ is contained within the complement $X \setminus C^0([0,1])$. Since the complement is open, $C^0([0,1])$ is closed!

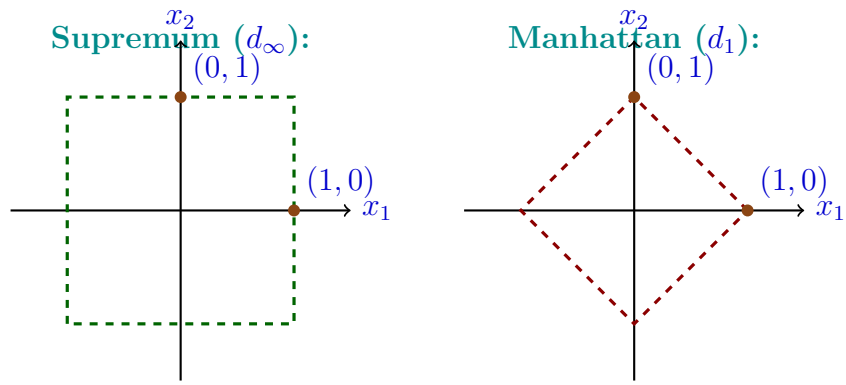
□

Friday

Warm-up. What is the shape of the open unit ball $B_1(0) \subseteq \mathbb{R}^2$ with:

- The Supremum metric?
- The Manhattan metric?

Answer.



Definition (Continuous functions)**.** For two metric spaces (X, d_X) and (Y, d_Y), a function $f : X \to Y$ is called *continuous* if one of the following, completely equivalent, conditions holds:

a) $(\varepsilon - \delta)$**-criterion:** $\forall x_0 \in X, \forall \varepsilon > 0\, \exists \delta > 0$ such that $\forall x \in X$:

$$d_X(x_0, x) < \delta \implies d_Y(f(x_0), f(x)) < \varepsilon$$

(If you are close enough to x_0 in the starting space, you are guaranteed to land close to $f(x_0)$ in the target space.)

b) **Sequential continuity:** For any sequence $(x_n)_{n=0}^{\infty}$ in X with a limit $x \in X$, it holds that:

$$\lim_{n\to\infty} f(x_n) = f(x)$$

(Continuous functions allow you to pull the limit inside the argument.)

c) **Topological continuity:** For all open sets $U \subseteq Y$, the preimage $f^{-1}(U) \subseteq X$ is *open*. (Continuous functions preserve the topological structure when mapping backwards.)

Example. Let $f : \mathbb{R} \to \mathbb{R}$, $x \mapsto x^2$ and let $(a, b) \subseteq \mathbb{R}$ be an interval. The preimage of this interval depends heavily on a and b:

- For $a < 0 < b$: $f^{-1}((a, b)) = (-\sqrt{b}, \sqrt{b})$. (This is an open interval!)
- For $a < b < 0$: $f^{-1}((a, b)) = \emptyset$. (The empty set is, as we proved, open!)
- For $0 < a < b$: $f^{-1}((a, b)) = (-\sqrt{b}, -\sqrt{a}) \cup (\sqrt{a}, \sqrt{b})$. (A union of open intervals is open!)

Why is it enough to check only these simple basis intervals?
Hint: The preimage perfectly respects unions: $f^{-1}(U_1 \cup U_2) = f^{-1}(U_1) \cup f^{-1}(U_2)$. Any open set on $\mathbb{R}$ can be constructed by patching together such intervals.

Question. Consider functions mapping between two metric spaces (X, d_X) and (Y, d_Y).

a) What happens if d_Y is the discrete metric? Is a standard function like $f : (\mathbb{R}, d_{std}) \to (\mathbb{R}, d_{discrete})$ continuous?

b) And conversely: What if the starting space d_X is discrete, and the target space d_Y is entirely arbitrary?

Answer. **a)** No, almost never. Take the identity function $f(z) = z$ as an example. In the discrete space Y, even a single point like $\{0\}$ is an open set! Let's look at its preimage: $\mathrm{Id}^{-1}(\{0\}) = \{0\}$. But a single point is definitely **not** open in the starting space with the standard metric d_{std}. The preimage of an open set is not open here, so the function is not continuous!

b) Yes, always! In a discrete space, every single subset is open. This means that no matter what open set $U \subseteq Y$ you pick, its preimage $f^{-1}(U)$ is guaranteed to be open in X (because *everything* is open there). Any function mapping out of a discrete space is automatically continuous.

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5. Completeness and Compactness

Definition (Cauchy Sequences & Complete Metric Spaces)**.** Let (X, d) be a metric space, and $(x_n)_{n=0}^{\infty}$ a sequence in X. We call (x_n) a **Cauchy sequence** if its terms get arbitrarily close to one another beyond a certain point:

$$\forall \varepsilon > 0 \, \exists N \in \mathbb{N} \text{ such that } d(x_n, x_m) < \varepsilon \quad \forall n, m \geq N$$

A metric space (X, d) is called a **complete metric space** if every Cauchy sequence in X actually converges to a real limit within X.

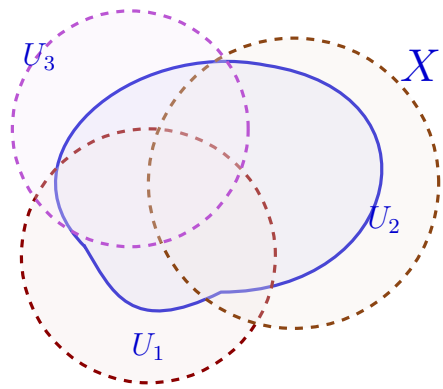
[Gemini-Notiz: Intuition for completeness: A Cauchy sequence desperately "wantst- to converge. A space is incomplete if it sabotages this—for instance, if the space has *„holes“* (like $\mathbb{Q}$) or is missing its boundary, meaning the sequence steers towards a limit that isn't actually there.]

Example.

- The Euclidean spaces $(\mathbb{R}^n, \langle \cdot, \cdot \rangle_{\text{eucl}})$ are complete.
- Any non-closed subset $U \subseteq \mathbb{R}^n$ and the rational numbers $\mathbb{Q}$ are *not* complete.
- The space of continuous functions $(C^0([0, 1]), \sup)$ is complete!

Definition (Compactness)**.** A metric space (X, d) is compact if every wild, infinite open cover can be reduced to a *finite* subcover.

Theorem. A metric space (X, d) is compact if and only if *every* sequence in X has a subsequence that converges to a point in X.



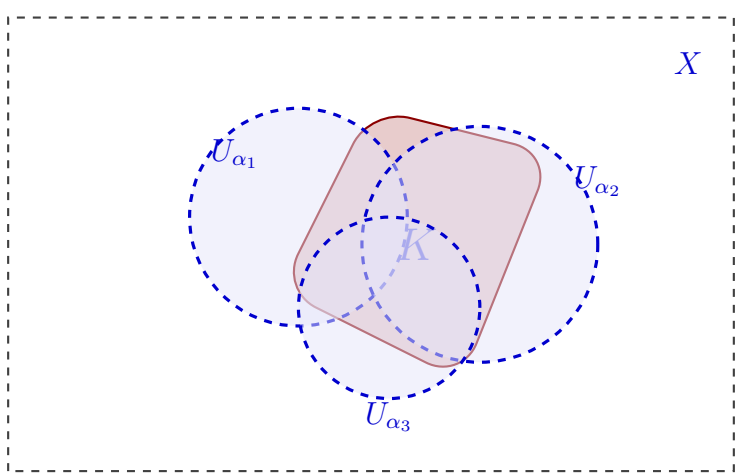
$U_1 \cup U_2 \cup U_3 = X$

Note: A subset K of a metric space (X, d) is compact if the subspace (K, d) is compact. This is strictly equivalent to the following prominent definition from the official script:

Definition 9.63 (Compactness)**.** Let (X, d) be a metric space. A subset $K \subseteq X$ is called compact if one (and therefore both) of the following equivalent conditions hold:

a) K is *„sequentially compact“*: every sequence $(x_n)_{n\in\mathbb{N}}$ in K has a subsequence that is convergent in K.

b) K is *„topologically compact“*: every family of open sets $\{U_i\}_{i\in I}$ that cover K has a *finite* subcover.



Important: The sets $U_i \subseteq X$ in the cover are always *open sets* with respect to the ambient

space (X, d).

Exercises. **a)** Show that every finite subset $\{x_1, \ldots, x_n\} \subseteq X$ is compact.

b) If a set K is compact, is it automatically complete?

Solution.
a) Let's use topological compactness: Let $\{U_\alpha\}_{\alpha \in I}$ be an arbitrary open cover of $\{x_1, \ldots, x_n\}$. Since these sets must cover all our points, we can simply pick *one* set U_{α_i} from the massive pool for each point x_i (from $i = 1$ to n) such that $x_i \in U_{\alpha_i}$. The collection $\{U_{\alpha_1}, \ldots, U_{\alpha_n}\}$ is then a subcover. Because it consists of only n (a finite number!) sets, we have found a finite subcover. Thus, the finite set is compact.

b) Yes, absolutely. Sequential compactness is perfect for this: Let $(x_n)_{n=0}^{\infty}$ be an arbitrary Cauchy sequence in X. Because K is (sequentially) compact, this sequence *must* have a convergent subsequence:

$$\lim_{i \to \infty} x_{n_i} = x \in X$$

Now a fundamental trick of Cauchy sequences comes into play: If a Cauchy sequence agrees to converge even for a fraction of its terms (the subsequence), it forcibly drags the rest of the sequence along with it!

Formally: Fix an $\varepsilon > 0$. Because it is a Cauchy sequence, all terms move within ε-distance of each other beyond some index N_1:

$$d(x_n, x_m) < \varepsilon \quad \forall n, m > N_1$$

Because the subsequence converges to x, its terms move within ε-distance of x beyond some index N_2:

$$d(x, x_{n_i}) < \varepsilon \quad \forall i > N_2$$

If we go far enough into the sequence (choosing $N := \max\{N_1, N_2\}$), we can estimate the distance of any arbitrary term x_k (with $k > N$) to the limit x. We just cleverly insert an element x_{n_k} of the convergent subsequence in between:

$$d(x, x_k) \leq d(x, x_{n_k}) + d(x_{n_k}, x_k) < 2\varepsilon$$

(Where of course $n_k \geq k > N$). Since we can make the distance 2ε arbitrarily small, the entire Cauchy sequence indeed converges to x. Thus, every compact space is always complete! $\square$

3. Compactness (Continued)

Theorem (Heine-Borel)**.** A subset $K \subseteq \mathbb{R}^n$ equipped with the *standard metric* is compact if and only if it is closed and bounded.

[Gemini-Notiz: The Heine-Borel theorem perfectly captures our intuitive visual understanding of a compact"(small, contained, boundary-inclusive) set in standard geometry. However, this beautiful equivalence strictly relies on the standard Euclidean space. We will now prove that Heine-Borel completely breaks down if we use a non-standard metric!]

Exercise. Let (X, d) be a metric space.

a) Show that $\tilde{d}(x, y) := \frac{d(x,y)}{1+d(x,y)}$ is also a valid metric on X.

b) Show that if $U \subseteq X$ is open with respect to d, then U is also open with respect to $\tilde{d}$. (Optional)

c) Show that for $X = \mathbb{R}^n$ with the metric $\tilde{d}(x, y) = \frac{|x-y|}{1+|x-y|}$, the entire space $\mathbb{R}^n \subseteq X$ is closed and bounded, but *not* compact.

Solution. **a)** Non-negativity, definiteness, and symmetry follow immediately from the properties of $d(x, y)$. For the triangle inequality, we want to show:

$$\frac{d(x,y)}{1+d(x,y)} \leq \frac{d(x,z)}{1+d(x,z)} + \frac{d(z,y)}{1+d(z,y)}$$

Let's denote $d_1 = d(x, y)$, $d_2 = d(x, z)$, and $d_3 = d(z, y)$. We cross-multiply to clear the denominators:

$$d_1(1+d_2)(1+d_3) \leq d_2(1+d_1)(1+d_3) + d_3(1+d_1)(1+d_2)$$

Expanding everything yields:

$$d_1 + d_1d_2 + d_1d_3 + d_1d_2d_3 \leq d_2 + d_1d_2 + d_2d_3 + d_1d_2d_3 + d_3 + d_1d_3 + d_2d_3 + d_1d_2d_3$$

Canceling the common terms (d_1d_2, d_1d_3, and one $d_1d_2d_3$) leaves us with:

$$d_1 \leq d_2 + 2d_2d_3 + d_3 + d_1d_2d_3$$

We know from the standard triangle inequality that $d_1 \leq d_2 + d_3$. Subtracting this known truth from both sides yields:

$$0 \leq 2d_2d_3 + d_1d_2d_3$$

Since distances are non-negative, this final inequality is trivially true! Thus, $\tilde{d}$ is a metric.

b) (Proof sketch) The function $t \mapsto \frac{t}{1+t}$ is strictly monotonically increasing for $t \geq 0$. Therefore, open balls in d can be directly mapped to open balls in $\tilde{d}$ by simply transforming the radius r into $\tilde{r} = \frac{r}{1+r}$.

c) Note that for any $t \geq 0$, the expression $\frac{t}{1+t}$ is strictly strictly less than 1. Therefore, for all $x, y \in \mathbb{R}^n$:

$$\tilde{d}(x, y) = \frac{|x - y|}{1 + |x - y|} < 1$$

This means the entire space $\mathbb{R}^n$ fits inside the open ball $\tilde{B}_1(0)$. Thus, $\mathbb{R}^n$ is **bounded** with respect to this metric! Also, the full space is trivially **closed**. However, consider the sequence $x_n = (n, 0, \ldots, 0)$. This sequence wanders off to infinity and has absolutely no convergent subsequence. Since it lacks sequential compactness, $\mathbb{R}^n$ is **not compact**, breaking Heine-Borel! $\square$

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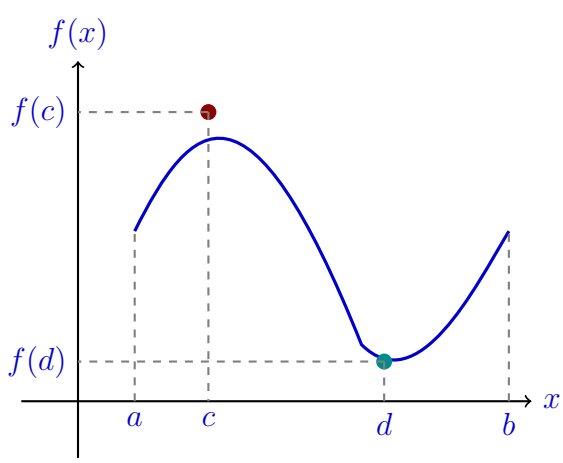
4. Why do we care about compactness?

Compactness is one of the most powerful properties a space can have. It grants us three massive mathematical guarantees:

a) **Preserved by continuous functions:** Let X, Y be metric spaces, and $f : X \to Y$ continuous. If $K \subseteq X$ is compact, then its image $f(K) \subseteq Y$ is guaranteed to be compact.

b) **Weierstrass Theorem / Extreme Value Theorem:** Let $K \subseteq X$ be a compact metric space, and $f : K \to \mathbb{R}$ a continuous function. Then f attains an absolute maximum and minimum within K. (In particular, f is bounded!)

c) **Uniform continuity:** Let X, Y be metric spaces, and $f : X \to Y$ continuous. If X is compact, then f is automatically *uniformly* continuous! (You can pick a global δ that works everywhere for a given ε.)



A continuous function on a compact interval $[a, b]$ always attains its absolute maximum (red) and absolute minimum (cyan).

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5. Connectedness

Definition (Disconnected metric space/subset)**.** A metric space (X, d) is **disconnected** if there exist subsets $U_1, U_2 \subseteq X$ which are:

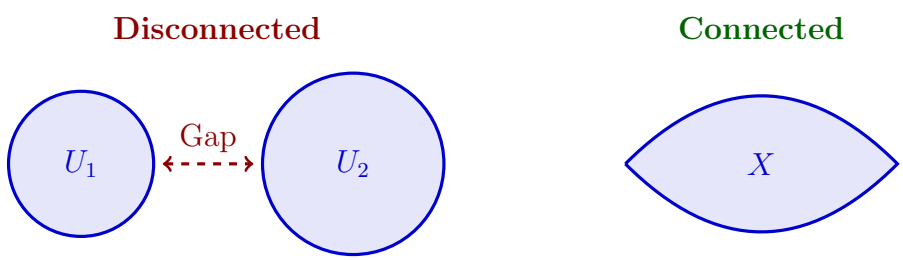
a) **open**

b) **non-empty** ($U_1 \neq \emptyset \neq U_2$)

c) **disjoint** ($U_1 \cap U_2 = \emptyset$)

such that $X = U_1 \cup U_2$. Similarly, a subset $A \subseteq X$ is disconnected if it can be split in this exact same way using the induced metric.
Connected is simply defined as the logical negation of disconnected!

[Gemini-Notiz: Intuitively, a space is disconnected if it consists of two or more completely isolated *„islands“*. Because proving connectedness directly is very hard, we almost always prove it by contradiction: We assume the space *is* disconnected (split into two islands U_1 and U_2) and show that this leads to a logical paradox.]



Theorem. Continuous functions preserve connected sets! Let X, Y be metric spaces, and $f : X \to Y$ continuous. If $A \subseteq X$ is connected, then its image $f(A)$ is strictly connected in Y.

Exercise. Are the following statements True or False?

a) The union of connected subsets with a common point is connected.

b) The intersection of connected subsets is connected.

c) The closure of a connected subset is connected.

Solution. **1. True.**
Proof: Let X be a metric space, $V_1, V_2 \subseteq X$ connected, and let $x_0 \in V_1 \cap V_2$ be their shared point. Assume by contradiction that their union is disconnected. Then $V_1 \cup V_2 = U_1 \cup U_2$, where $U_1, U_2 \subseteq X$ are open, non-empty, and disjoint. Without loss of generality, assume the shared point x_0 lies in U_1. Because x_0 is in both V_1 and V_2, it follows that $U_1 \cap V_1 \neq \emptyset$ and $U_1 \cap V_2 \neq \emptyset$. Since V_1 and V_2 are individually connected, they cannot be split across U_1 and U_2. Since they both share a point with U_1, they must both lie *entirely* inside U_1. Thus $V_1 \cap U_2 = \emptyset$ and $V_2 \cap U_2 = \emptyset$. This forces $U_2 = \emptyset$, which violates the definition of disconnectedness! Contradiction.

2. False.
Counterexample: Let $X = \mathbb{R}^2$. Let $U = [-1, 1] \times \{0\}$ (a horizontal line segment) and $V = S^1$ (the unit circle). Both are perfectly connected. However, their intersection is exactly two isolated points: $U \cap V = \{(-1, 0), (1, 0)\}$, which is clearly disconnected!

3. True.
Proof: Let A be connected with closure $\bar{A}$. Suppose $\bar{A}$ is disconnected, meaning there exist disjoint open sets U_1, U_2 covering $\bar{A}$. Without loss of generality, let $A \cap U_1 \neq \emptyset$. Since A is connected and covered by $U_1 \cup U_2$, it must lie entirely in one of them, meaning $A \cap U_2 = \emptyset$. However, U_2 is an open set. If an open set contains no points of A, it cannot possibly contain any boundary points of A either! Thus $U_2 \cap \bar{A} = \emptyset$, meaning U_2 is empty. Contradiction. $\square$

---

6. Path Connectedness

Definition (Paths and Path-connectedness)**.** Let (X, d) be a metric space.

- A **path** is a continuous function $\gamma : [0, 1] \to X$, where $[0, 1]$ carries the standard metric.
- A space (or subset) X is **path-connected** if for any two points $x, y \in X$, there exists a path γ such that $\gamma(0) = x$ and $\gamma(1) = y$.

Proposition 1.
- **Path-connected** $\implies$ **connected.** (This is always true!)
- For *open* subsets of Euclidean space $\mathbb{R}^n$: **connected** $\iff$ **path-connected.**

Counterexample: The Topologist's Sine Curve

The reverse implication (connected $\implies$ path-connected) is **false** for non-open sets! Consider the set:

$$S := \left\{ \left(t, \sin\left(\frac{1}{t}\right) \right) : t > 0 \right\} \cup \{(0, 0)\} \subseteq \mathbb{R}^2$$

S is **connected** (because the sine wave part is connected, and we proved earlier that adding boundary points via the closure maintains connectedness). However, S is **not path-connected**!

[Gemini-Notiz: Why can't we draw a path? As t approaches 0, the frequency of the sine wave goes to infinity. It oscillates up and down infinitely fast between -1 and 1. If you try to draw a path from the origin $(0,0)$ to the rest of the curve, your pen would have to cross the x-axis an infinite number of times in a finite amount of time, shattering continuity!]

Solution. Assume by contradiction that S is path-connected. Then there exists a continuous path $\gamma : [0,1] \to S$ starting at the origin $\gamma(0) = (0,0)$ and ending at some point on the curve, say $\gamma(1) = (1, \sin(1))$. We can write the path as two coordinate functions $\gamma(t) = (x(t), y(t))$. Since γ is continuous, $x(t)$ and $y(t)$ must be continuous.

By the Intermediate Value Theorem, since $x(0) = 0$ and $x(1) = 1$, the function $x(t)$ must hit every x-coordinate in between. Let's look at the peaks of the sine wave, which occur at x-coordinates $\frac{1}{2\pi n + \pi/2}$. We can construct a sequence of times $(a_n)_{n=0}^{\infty}$ in $[0,1]$ such that:

$$x(a_n) = \frac{1}{2\pi n + \pi/2}$$

Notice that the y-coordinate at all these peaks is exactly $y(a_n) = 1$. As $n \to \infty$, the x-coordinate goes to 0, so $a_n \to 0$. But look at the limits:

$$\lim_{n\to\infty} \gamma(a_n) = (0,1)$$

However, sequential continuity demands that $\lim \gamma(a_n) = \gamma(0) = (0,0)$. $(0,1) \neq (0,0)$, which is a massive contradiction! Thus, no such continuous path can exist. $\square$

---

7. Inner Product and Norm

Definition (Inner Product)**.** Let V be a real vector space. An **inner product** on V is a map $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{R}$ satisfying:

a) **Bilinearity:** $\langle \alpha u + \beta v, w \rangle = \alpha \langle u, w \rangle + \beta \langle v, w \rangle$ (and equivalently in the second variable)

b) **Symmetry:** $\langle u, v \rangle = \langle v, u \rangle$

c) **Definiteness:** $\langle u, u \rangle \geq 0$, and $\langle u, u \rangle = 0 \iff u = 0$

Definition (Norm)**.** A **norm** on V is a map $\|\cdot\| : V \to [0, \infty)$ satisfying:

a) **Definiteness:** $\|v\| \geq 0$, and $\|v\| = 0 \iff v = 0$

b) **Homogeneity:** $\|\alpha v\| = |\alpha|\|v\|$

c) **Triangle Inequality:** $\|u + v\| \leq \|u\| + \|v\|$

Geometric Meaning and Projections

Every inner product naturally defines a norm via $\|v\| := \sqrt{\langle v, v\rangle}$. In $\mathbb{R}^2$, the standard inner product geometrically yields the angle θ between two vectors:

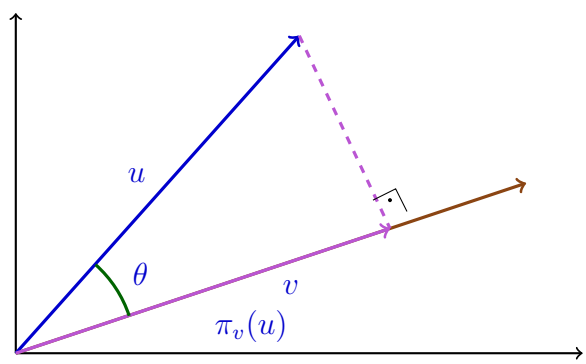
$$\langle x, y\rangle = \|x\|\|y\| \cos\theta$$

In a general inner product space, this formula actually serves as the *definition* of an angle!

This allows us to define the orthogonal projection of a vector u onto a vector v:

$$\pi_v(u) = \frac{\langle u, v\rangle}{\|v\|^2} v$$

In words, $\pi_v(u)$ is the ßhadowöf u pointing exactly along the direction of v.



The Cauchy-Schwarz Inequality

If we take the norm of the projection, we get:

$$\|\pi_v(u)\| = \|u\||\cos\theta| = \frac{|\langle u, v\rangle|}{\|v\|}$$

Because the ßhadowöf a vector can never be longer than the vector itself ($|\cos\theta| \leq 1$), it must hold that $\|\pi_v(u)\| \leq \|u\|$. Moving the denominator over yields the legendary Cauchy-Schwarz Inequality:

$$|\langle u, v\rangle| \leq \|u\|\|v\|$$

[Gemini-Notiz: Hint for a rigorous algebraic proof: Use the definiteness property! We know that the length of the *„perpendicular part“* of the vector must be ≥ 0. So, expand the inner product of the perpendicular part with itself: $\langle u - \pi_v(u), u - \pi_v(u)\rangle \geq 0$. Expanding this term will elegantly spit out the Cauchy-Schwarz inequality!]

4. The Differential

To understand how derivatives work in multiple dimensions, let's first recall the 1-dimensional case from Analysis 1. For a function $f : \mathbb{R} \to \mathbb{R}$ that is differentiable at $x_0 \in \mathbb{R}$, we can write its linear approximation as:

$$f(x_0 + h) = f(x_0) + f'(x_0)h + o(h)$$

This is perfectly equivalent to the classic limit definition:

$$\lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h} = f'(x_0) \quad \iff \quad \lim_{h \to 0} \frac{|f(x_0 + h) - f(x_0) - f'(x_0)h|}{|h|} = 0$$

Now, let's upgrade to a multivariable function $F : \mathbb{R}^n \to \mathbb{R}^m$. We can no longer just multiply by a scalar slope $f'(x_0)$. Instead, the derivative must be a **linear map** that transforms an n-dimensional step h into an m-dimensional change in output!

Definition (Differentiability)**.** We say that $F : \mathbb{R}^n \to \mathbb{R}^m$ (or $F : U \to \mathbb{R}^m$ for an open subset $U \subseteq \mathbb{R}^n$) is **differentiable** at x_0 if there exists a linear map $DF_{x_0} : \mathbb{R}^n \to \mathbb{R}^m$ such that:

$$F(x_0 + h) = F(x_0) + DF_{x_0}(h) + o(|h|)$$

Or, written as a limit:

$$\lim_{h \to 0} \frac{|F(x_0 + h) - F(x_0) - DF_{x_0}(h)|}{|h|} = 0$$

[Gemini-Notiz: Notice the absolute value bars in the denominator! We divide by the *length* of the vector h. If you look closely at the term $\frac{F(x_0+h)-F(x_0)}{|h|}$, you can see that as $h \to 0$, this evaluates the rate of change along a specific direction $\frac{h}{|h|}$. This is intimately connected to the directional derivative!]

The Jacobian Matrix

Since $DF_{x_0} : \mathbb{R}^n \to \mathbb{R}^m$ is a linear map, it can be represented by an $m \times n$ matrix—provided we fix a basis. In the standard basis of $\mathbb{R}^n$, this matrix is the famous **Jacobian matrix**:

$$DF_x \approx JF(x) = \begin{pmatrix} \frac{\partial F_1}{\partial x_1} & \cdots & \frac{\partial F_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial F_m}{\partial x_1} & \cdots & \frac{\partial F_m}{\partial x_n} \end{pmatrix}$$

Here, $F(x) = (F_1(x), \dots, F_m(x))^T$, and the i-th partial derivative is defined as taking the 1D derivative along the i-th standard basis vector e_i:

$$\partial_i F(x) = \lim_{s \to 0} \frac{F(x + se_i) - F(x)}{s}$$

If F is fully differentiable, the partial derivatives are simply the columns of the Jacobian, and we can compute them by applying the differential to the basis vectors:

$$\partial_i F(x) = DF_x(e_i)$$

[Gemini-Warnung: Crucial Trap: Partial derivatives can exist even if F is NOT differentiable! Partial derivatives only check the grid lines (x-axis, y-axis). A function could be perfectly smooth along the axes but have a massive tear or crease diagonally. Full differentiability requires the linear approximation DF_x to work from *all* approach directions simultaneously.]

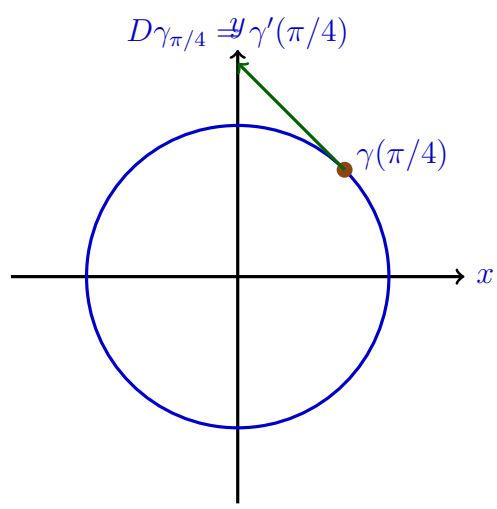
Examples of Differentials

Example (1. A Curve in 2D Space)**.** Consider a path $\gamma : \mathbb{R} \to \mathbb{R}^2, t \mapsto \binom{\cos t}{\sin t}$.
Let's find the differential at $t = \frac{\pi}{4}$. For a curve, the Jacobian is just a column vector (the velocity vector):

$$\gamma'\left(\frac{\pi}{4}\right) := D\gamma_{\pi/4} = \begin{pmatrix} \partial_t \gamma_1(\pi/4) \\ \partial_t \gamma_2(\pi/4) \end{pmatrix} = \begin{pmatrix} -\sin(\pi/4) \\ \cos(\pi/4) \end{pmatrix} = \begin{pmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$$

Using the approximation formula $F(x_0 + h) = F(x_0) + DF_{x_0}(h) + o(|h|)$:

$$\gamma\left(\frac{\pi}{4} + h\right) = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix} + \frac{h}{\sqrt{2}}\begin{pmatrix} -1 \\ 1 \end{pmatrix} + o(|h|)$$



Notice how the column vectors of the Jacobian are the partial derivatives. If we have a general direction vector $v = (v_1, \ldots, v_n)^T$, the directional derivative is just the matrix-vector product: $\partial_v F(x) = DF_x(v) = JF(x)v = \sum_{i=1}^n v_i \frac{\partial F}{\partial x_i}$.

Example (2. A Scalar Field)**.** Consider $f : \mathbb{R}^3 \to \mathbb{R}$, given by $(x, y, z) \mapsto 2x^2+y^2+3z^2-2xyz$. What is the differential $df(x, y, z)$? Since the output is 1D, the Jacobian is a row vector:

$$Jf(x, y, z) = (4x - 2yz, \quad 2y - 2xz, \quad 6z - 2xy)$$

Notice that this is exactly the transpose of the gradient!

$$Jf(x, y, z) = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\right) = (\nabla f)^T$$

---

5. The Chain Rule

For differentiable functions $f : U \subseteq \mathbb{R}^n \to \mathbb{R}^m$ and $g : V \subseteq \mathbb{R}^m \to \mathbb{R}^l$, the differential of the composition is the composition of the differentials:

$$D(g \circ f)(x) = Dg(f(x)) \cdot Df(x)$$

Since linear maps compose via matrix multiplication, this translates directly to their Jacobians:

$$J(g \circ f)(x) = Jg(f(x)) \cdot Jf(x)$$

Exercise. Compute $J(g \circ f)$ for the following functions:

- $f : \mathbb{R}^3 \to \mathbb{R}^2, \quad x \mapsto (x_1x_2,\ x_2 + x_3)^T$
- $g : \mathbb{R}^2 \to \mathbb{R}^2, \quad y \mapsto (e^{y_1},\ y_1y_2)^T$

Solution. First, compute the individual Jacobians:

$$Jf(x) = \begin{pmatrix} x_2 & x_1 & 0 \\ 0 & 1 & 1 \end{pmatrix}, \quad Jg(y) = \begin{pmatrix} e^{y_1} & 0 \\ y_2 & y_1 \end{pmatrix}$$

Before multiplying, we must evaluate Jg at the point $y = f(x)$, so $y_1 = x_1x_2$ and $y_2 = x_2+x_3$:

$$Jg(f(x)) = \begin{pmatrix} e^{x_1x_2} & 0 \\ x_2 + x_3 & x_1x_2 \end{pmatrix}$$

Now, multiply the matrices:

$$\begin{aligned} J(g \circ f)(x) &= \begin{pmatrix} e^{x_1 x_2} & 0 \\ x_2 + x_3 & x_1 x_2 \end{pmatrix} \begin{pmatrix} x_2 & x_1 & 0 \\ 0 & 1 & 1 \end{pmatrix} \\ &= \begin{pmatrix} x_2 e^{x_1 x_2} & x_1 e^{x_1 x_2} & 0 \\ x_2(x_2 + x_3) & x_1(x_2 + x_3) + x_1 x_2 & x_1 x_2 \end{pmatrix} \end{aligned} \tag{2}$$

□

Alternative: Directional Derivatives

Sometimes the differential is not easily expressed as a Jacobian matrix (e.g., when the vector space is the space of matrices). In such cases, it is extremely convenient to use the formula for directional derivatives:

$$DF_{x_0}(v) = \partial_v F(x_0) = \frac{d}{ds}\bigg|_{s=0} F(x_0 + sv)$$

Note: The directional derivative $\partial_v F(x_0)$ can exist even if F is not fully differentiable at x_0. However, if DF_{x_0} exists, this formula is always valid and often much faster to compute!

Exercise (Matrix Inner Products)**.** We identify the space of real-valued matrices $\mathbb{R}^{n\times n}$ with $\mathbb{R}^{n^2}$ using the standard inner product $\langle A, B\rangle = \mathrm{Tr}(A^T B)$. Consider the matrix function:

$$F : \mathbb{R}^{n\times n} \to \mathbb{R}^{n\times n}, \quad A \mapsto A^T A$$

Compute the differential at the identity matrix $Id \in \mathbb{R}^{n\times n}$, applied to a direction matrix $X \in \mathbb{R}^{n\times n}$:

$$DF_{Id}(X) = \partial_X F(Id)$$

Solution. Using the directional derivative trick, we evaluate the curve $Id + sX$ as it passes through Id:

$$\begin{aligned} DF_{Id}(X) &= \frac{d}{ds}\bigg|_{s=0} F(Id + sX) \\ &= \frac{d}{ds}\bigg|_{s=0} (Id + sX)^T (Id + sX) \\ &= \frac{d}{ds}\bigg|_{s=0} [Id^T Id + sX^T Id + sId^T X + s^2 X^T X] \\ &= \frac{d}{ds}\bigg|_{s=0} [Id + s(X^T + X) + s^2 X^T X] \end{aligned} \tag{3}$$

Taking the derivative with respect to s and plugging in $s = 0$ eliminates the constant term and the s^2 term, leaving exactly the linear part:

$$DF_{Id}(X) = X^T + X$$

□

6. Taylor Expansions in $\mathbb{R}^n$

In the lecture, you saw the terrifying full formula for a function $f \in C^k(U, \mathbb{R})$:

$$f(\bar{x} + h) = \sum_{|\alpha|=0}^{k} \partial^\alpha f(\bar{x}) \frac{h^\alpha}{\alpha!} + o(|h|^k)$$

where $\alpha \in \mathbb{N}^n$ is a *„multi-index“*.

[Gemini-Notiz: What on earth is a multi-index? It's just a compact way to track all the crazy mixed partial derivatives! $\alpha = (\alpha_1, \ldots, \alpha_n)$ tells you how many times to differentiate with respect to each variable. The sum $|\alpha| = \alpha_1 + \cdots + \alpha_n$ is the total order of the derivative. For example, $\partial^{(2,1)} = \partial_x^2 \partial_y$.]

Exercise. Let $f \in C^3(\mathbb{R}^2, \mathbb{R})$. Write out the maximal Taylor approximation (up to 3rd order) using multi-indices.

Solution. Let's list all possible multi-indices $\alpha = (\alpha_1, \alpha_2)$ up to order 3:

- $|\alpha| = 0$: $(0, 0)$
- $|\alpha| = 1$: $(1, 0), (0, 1)$
- $|\alpha| = 2$: $(2, 0), (1, 1), (0, 2)$
- $|\alpha| = 3$: $(3, 0), (2, 1), (1, 2), (0, 3)$

Recall the factorial rule: $\alpha! = \alpha_1! \alpha_2!$. For example, $(2, 1)! = 2! \cdot 1! = 2$. Plugging this into the formula yields an explosion of terms:

$$\begin{aligned} f(\bar{x} + h) = f(\bar{x}) &+ \partial_1 f(\bar{x}) h_1 + \partial_2 f(\bar{x}) h_2 \\ &+ \frac{1}{2} \partial_1^2 f(\bar{x}) h_1^2 + \partial_1 \partial_2 f(\bar{x}) h_1 h_2 + \frac{1}{2} \partial_2^2 f(\bar{x}) h_2^2 \\ &+ \frac{1}{6} \partial_1^3 f(\bar{x}) h_1^3 + \frac{1}{2} \partial_1^2 \partial_2 f(\bar{x}) h_1^2 h_2 + \frac{1}{2} \partial_1 \partial_2^2 f(\bar{x}) h_1 h_2^2 + \frac{1}{6} \partial_2^3 f(\bar{x}) h_2^3 \\ &+ o(|h|^3) \end{aligned} \tag{4}$$

□

The Working Formula and Substitution Trick

Because the multi-index formula is a nightmare to write out, we often rewrite it using the gradient operator as a polynomial:

$$f(\bar{x} + h) = \sum_{l=0}^{k} \frac{1}{l!} \left(\sum_{m=1}^{n} h_m \partial_m \right)^l f(\bar{x}) + o(|h|^k)$$

However, the absolute best way to compute Taylor series is often to **cheat** using known 1D series from Analysis 1!

Exercise. Compute the Taylor expansion of $f(x, y) = (x + y^2)e^{-x^2-y^2}$ around $(0, 0)$ up to third order.

Solution. **Cumbersome solution:** Calculate all partial derivatives up to the 3rd order and plug them into the massive formula above. (Please never do this in an exam!)

The Easy Solution: Use the substitution and multiplication rules from Analysis 1. We know the 1D Taylor series for $e^z = 1 + z + \frac{z^2}{2} + \dots$. Substitute $z = -x^2 - y^2$:

$$e^{-x^2-y^2} = 1 + (-x^2 - y^2) + o(|(x, y)|^3)$$

Now, multiply this by the front bracket, dropping any terms that result in a power higher than 3:

$$\begin{aligned} f(x, y) &= (x + y^2)\left(1 - x^2 - y^2 + o(|(x, y)|^3)\right) \\ &= x + y^2 - x^3 - xy^2 + o(|(x, y)|^3) \end{aligned} \tag{5}$$

And we are done in two lines of algebra! □

Remark. The 2nd order Taylor approximation of $f \in C^2(\mathbb{R}^n, \mathbb{R})$ around $x_0 \in \mathbb{R}^n$ can be expressed beautifully using matrices:

$$f(x_0 + h) = f(x_0) + \nabla f(x_0)^T h + \frac{1}{2} h^T \mathscr{H} f(x_0) h + o(|h|^2)$$

where $\mathscr{H} f(x_0) = (\partial_i \partial_j f(x_0))_{i,j}$ is the symmetric **Hessian matrix** of f. (Notice how perfectly this mirrors the 1D formula: $f(x_0) + f'h + \frac{1}{2} f'' h^2$!)

5. Optimization

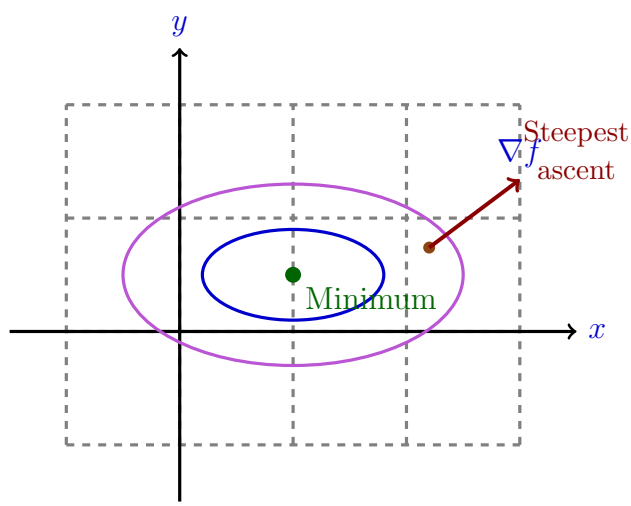
Let's look at finding the extreme values of a function $f : U \subseteq \mathbb{R}^n \to \mathbb{R}$.

Definition (Local Extrema and Critical Points).

- f has a **local minimum** at $x_0 \in U$ if there exists a ball $B_\varepsilon(x_0)$ such that $\forall x \in B_\varepsilon(x_0) \cap U$, we have $f(x) \geq f(x_0)$. (Analogous for local maximum with $\leq$).
- Extrema are **strict** if the inequality is strict ($<$ or $>$).
- A **critical point** of $f \in C^1(U, \mathbb{R})$ is any point x_0 where the gradient vanishes: $\nabla f(x_0) = 0$.

Proposition 2. If $f \in C^1(U, \mathbb{R})$ has a local extremum at an *interior* point $x_0 \in \operatorname{int}(U)$, then x_0 must be a critical point ($\nabla f(x_0) = 0$).

[Gemini-Notiz: Geometrically, the gradient $\nabla f(x_0)$ always points in the direction of the *„steepest ascent“*. If you are at a local minimum (the bottom of a valley) or a local maximum (the peak of a mountain), there is no *„uphill“* direction anymore. You are standing perfectly flat. Therefore, the gradient vector must be the zero vector!]



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6. Constrained Optimization (Lagrange)

Suppose we want to find the hottest place on the surface of the Earth. In theory, temperature is a scalar field defined everywhere in space: $T \in C^\infty(\mathbb{R}^3, (0, \infty))$. We model the Earth's surface as the unit sphere $S^2 := \{x \in \mathbb{R}^3 : |x|^2 = 1\}$.

We want to find the local maxima of the *restricted* function $T|_{S^2}$. Crucially, these are **not necessarily** critical points of the global function T!

[Gemini-Notiz: Imagine standing on the equator. The true gradient of the temperature ∇T might point straight up into the sun (off the surface of the Earth). However, since you are constrained to walk only *along* the surface, you only care about the *„shadow“* of that gradient on the ground. A point is a constrained extremum if the true gradient points entirely off the surface, meaning it is perfectly perpendicular to the ground!]

Since our constraint surface S^2 is the level set of a function $g(x) = |x|^2 - 1 = 0$, we know that $\nabla g(x)$ is everywhere perpendicular to the surface. Thus, at a constrained extremum p, the temperature gradient $\nabla T(p)$ must be perfectly parallel to the constraint gradient $\nabla g(p)$.

Proposition 3 (Lagrange Multipliers)**.** Suppose $f : U \subseteq \mathbb{R}^n \to \mathbb{R}$ and $g = (g_1, \ldots, g_k) : U \to \mathbb{R}^k$ are C^1. If the restricted function $f|_{g^{-1}\{0\}}$ has a local extremum at x_0, then the gradients $\{\nabla f(x_0), \nabla g_1(x_0), \ldots, \nabla g_k(x_0)\}$ are linearly dependent.

Assuming the constraint gradients are independent, this means $\nabla f(x_0)$ is a linear combination of the constraint gradients. We can find these points by looking for critical points of the **Lagrangian**:

$$L(x, \lambda) := f(x) - \sum_{i=1}^{k} \lambda_i g_i(x)$$

We force $\nabla L(x, \lambda) = 0$, which yields the system:

$$\begin{aligned} \nabla f(x) - \lambda \cdot \nabla g(x) &= 0 \quad \text{(Parallel gradients)} \\ g_j(x) &= 0 \quad \text{(You are actually on the surface)} \end{aligned} \tag{6}$$

Exercise. Find the local extrema of $f(x, y, z) = xyz$ on the sphere $K = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 \leq 1\}$.

Solution. First, we check the **interior** ($x^2 + y^2 + z^2 < 1$) for standard critical points:

$$\nabla f(x, y, z) = (yz, xz, xy) = 0 \implies \text{at least two variables must be 0.}$$

For example, $(0, 0, 0)$, or $(x, 0, 0)$. All these yield $f = 0$.

Next, we check the **boundary** using Lagrange multipliers. The constraint is $g(x, y, z) = x^2 + y^2 + z^2 - 1 = 0$. The Lagrangian is:

$$L(x, y, z, \lambda) = xyz - \lambda(x^2 + y^2 + z^2 - 1)$$

Taking the partial derivatives and setting them to zero gives our system:

a) $\partial_x L = yz - 2\lambda x = 0 \implies yz = 2\lambda x$

b) $\partial_y L = xz - 2\lambda y = 0 \implies xz = 2\lambda y$

c) $\partial_z L = xy - 2\lambda z = 0 \implies xy = 2\lambda z$

d) $\partial_\lambda L = 1 - (x^2 + y^2 + z^2) = 0 \implies x^2 + y^2 + z^2 = 1$

[Gemini-Notiz: We want to divide by x, y, and z to isolate 2λ. Is that allowed? If $x = 0$, equation (1) forces $y = 0$ or $z = 0$. If $x = 0$ and $y = 0$, equation (4) forces $z = \pm 1$. The points $(0, 0, \pm 1)$ yield $f = 0$, which we already know are not strict extrema since we can easily find positive/negative values elsewhere. Thus, we can safely assume $x, y, z \neq 0$ for the true extrema and divide!]

Assuming non-zero coordinates, we isolate 2λ:

$$2\lambda = \frac{yz}{x} = \frac{xz}{y} = \frac{xy}{z}$$

Multiplying the terms pairwise yields:

$$y^2 z^2 = x^2 z^2 = x^2 y^2 \implies x^2 = y^2 = z^2$$

Plugging this symmetry into the constraint (4):

$$3x^2 = 1 \implies x = \pm \frac{1}{\sqrt{3}}$$

This gives $2^3 = 8$ solutions on the sphere:

$$(x, y, z) = \left(\frac{\alpha}{\sqrt{3}}, \frac{\beta}{\sqrt{3}}, \frac{\gamma}{\sqrt{3}} \right) \quad \text{where } \alpha, \beta, \gamma \in \{-1, 1\}$$

□

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7. The Hessian Test

Just like the second derivative test in Analysis 1, we can classify critical points in $\mathbb{R}^n$ using the second derivatives. The **Hessian matrix** of $f \in C^2(U, \mathbb{R})$ is:

$$\mathscr{H} f(x) = (\partial_i \partial_j f)_{i,j=1,\dots,n}$$

By Schwarz's Lemma, this matrix is perfectly symmetric.

Definition (Definiteness of a Symmetric Matrix)**.** Let $A \in \mathbb{R}^{n \times n}$ be symmetric. It is called:

- **Positive definite** ($A > 0$)**:** $v^T A v > 0 \quad \forall v \neq 0$
- **Negative definite** ($A < 0$)**:** $v^T A v < 0 \quad \forall v \neq 0$
- **Indefinite:** $\exists v, u$ such that $v^T A v > 0$ and $u^T A u < 0$

If x_0 is a critical point ($\nabla f(x_0) = 0$), then:

- $\mathscr{H} f(x_0)$ is positive definite $\implies$ **Local Minimum**
- $\mathscr{H} f(x_0)$ is negative definite $\implies$ **Local Maximum**
- $\mathscr{H} f(x_0)$ is indefinite $\implies$ **Saddle Point**

How to determine Definiteness?

Let $\lambda_1, \ldots, \lambda_n$ be the eigenvalues of A. Recall that $\det(A) = \prod \lambda_i$ and $\mathrm{Tr}(A) = \sum \lambda_i$.

- **For $n = 2$:**
  - $\det(A) < 0 \implies$ Indefinite
  - $\det(A) > 0$ and $\mathrm{Tr}(A) > 0 \implies$ Positive definite
  - $\det(A) > 0$ and $\mathrm{Tr}(A) < 0 \implies$ Negative definite
- **For $n \geq 3$ (The Hurwitz Criterion):** Check the leading principal minors (the determinants of the upper-left submatrices $A_1, A_2, \ldots, A_n$).
  - $A_1 > 0, A_2 > 0, A_3 > 0 \implies$ Positive definite
  - $A_1 < 0, A_2 > 0, A_3 < 0 \implies$ Negative definite (alternating signs!)
  - Anything else (excluding zeros) $\implies$ Indefinite

Example (Testing Matrices). **1.** $A = \begin{pmatrix} 2 & 2 \\ 2 & 9 \end{pmatrix}$. $\det = 18 - 4 = 14 > 0$. $\mathrm{Tr} = 11 > 0 \implies$ Positive definite.

2. $A = \begin{pmatrix} 4 & 2 \\ 2 & -4 \end{pmatrix}$. $\det = -16 - 4 = -20 < 0 \implies$ Indefinite.

[Gemini-Warnung: The handwritten notes contained a matrix that was not symmetric ($\begin{smallmatrix} -5 & 0 & 0 \\ 0 & -4 & -7 \\ 3 & -2 & -4 \end{smallmatrix}$). The Hurwitz criterion strictly applies *only* to symmetric matrices! Let's look at the correct symmetric matrix the TA meant to evaluate:]

3. $A = \begin{pmatrix} -5 & 0 & 0 \\ 0 & -4 & -2 \\ 0 & -2 & -4 \end{pmatrix}$. Let's check the Hurwitz minors:

$$\begin{aligned} A_1 &= -5 \quad (< 0) \\ A_2 &= \det \begin{pmatrix} -5 & 0 \\ 0 & -4 \end{pmatrix} = 20 \quad (> 0) \\ A_3 &= \det(A) = -5 \cdot (16 - 4) = -60 \quad (< 0) \end{aligned} \tag{7}$$

The signs alternate $(-, +, -)$, so this is Negative definite.

4. $A = \begin{pmatrix} -1 & 0 & -4 \\ 0 & 2 & -2 \\ -4 & -2 & 22 \end{pmatrix}$. $A_1 = -1$ and $A_2 = \det \begin{pmatrix} -1 & 0 \\ 0 & 2 \end{pmatrix} = -2$. Since the signs are $(-, -, \dots)$, it violates both the positive and negative definite patterns $\implies$ Indefinite.

Exercise. Find the critical points of $f_\alpha(x, y) = x^3 - y^3 + 3\alpha xy$ for a constant $\alpha \in \mathbb{R} \setminus \{0\}$, and classify them (min/max/saddle).

Solution. First, find the critical points by setting the gradient to zero:

$$\begin{aligned} \partial_x f &= 3x^2 + 3\alpha y = 0 \implies y = -\frac{x^2}{\alpha} \\ \partial_y f &= -3y^2 + 3\alpha x = 0 \end{aligned} \tag{8}$$

Substitute (1) into (2):

$$-3\left(-\frac{x^2}{\alpha}\right)^2 + 3\alpha x = 0 \implies -3\frac{x^4}{\alpha^2} + 3\alpha x = 0$$

Divide by 3 and factor out x:

$$x\left(\alpha - \frac{x^3}{\alpha^2}\right) = 0$$

This yields two cases:

- $x = 0 \implies y = 0$. Point: $(0, 0)$.
- $x^3 = \alpha^3 \implies x = \alpha \implies y = -\alpha$. Point: $(\alpha, -\alpha)$.

Now, compute the Hessian matrix:

$$\mathscr{H} f(x, y) = \begin{pmatrix} 6x & 3\alpha \\ 3\alpha & -6y \end{pmatrix}$$

Test point 1: $(0, 0)$

$$\mathscr{H} f(0, 0) = \begin{pmatrix} 0 & 3\alpha \\ 3\alpha & 0 \end{pmatrix}$$

$\det = -9\alpha^2 < 0 \implies$ **Saddle Point**.

Test point 2: $(\alpha, -\alpha)$

$$\mathscr{H} f(\alpha, -\alpha) = \begin{pmatrix} 6\alpha & 3\alpha \\ 3\alpha & 6\alpha \end{pmatrix} = 3\alpha \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

The determinant is $36\alpha^2 - 9\alpha^2 = 27\alpha^2 > 0$. The trace is 12α.

- If $\alpha > 0$: Trace is positive $\implies$ Positive definite $\implies$ **Local Minimum**.
- If $\alpha < 0$: Trace is negative $\implies$ Negative definite $\implies$ **Local Maximum**.

$\square$

6. Convexity

Definition (Convex Sets and Functions).

- A subset $A \subseteq \mathbb{R}^n$ is **convex** if for any $x, y \in A$ and any $t \in [0,1]$, the point $p = (1-t)x + ty$ is also an element of A. (The straight line segment connecting any two points in the set lies entirely within the set.)
- A function $f : A \to \mathbb{R}$ is **convex** if its domain A is a convex set, and for all $x, y \in A$ and $t \in [0,1]$:
$$f((1-t)x + ty) \leq (1-t)f(x) + tf(y)$$

[Gemini-Notiz: The functional inequality simply states that if you draw a straight secant line between two points on the graph of f, the actual graph of the function will always hang *below* (or exactly on) that secant line. It looks like a bowl.]

Proposition 4. Let $U \subseteq \mathbb{R}^n$ be open and convex, and $f \in C^2(U)$. The following statements are perfectly equivalent:

a) f is a convex function.

b) The Hessian matrix $\mathscr{H}f(x)$ is positive semi-definite (non-negative definite) for all $x \in U$.

c) $f(y) - f(x) \geq Df_x(y-x)$ for all $x, y \in U$. (The graph of the function always lies above any of its tangent planes.)

Example. The n-dimensional parabola $p : \mathbb{R}^n \to \mathbb{R}, x \mapsto |x|^2 = \sum_{i=1}^n x_i^2$ is convex. Taking the second derivatives, we get $\partial_i \partial_j p(x) = 2\delta_{ij}$ (which is 2 if $i = j$, and 0 otherwise). Thus, the Hessian is strictly diagonal:
$$\mathscr{H}p(x) = 2I_n$$
Every eigenvalue of this matrix is $2 > 0$, so the matrix is strictly positive definite. This immediately implies the function is convex!

Exercise. Let $U \subseteq \mathbb{R}^n$ be open and convex, and $f \in C^1(U)$.

a) Show that $f : U \to \mathbb{R}$ is convex if and only if the 1-dimensional slice functions $f_{x,y} : [0,1] \to \mathbb{R}, s \mapsto f(x + s(y-x))$ are convex for all $x, y \in U$.

b) Use this to prove the tangent plane property: f convex $\iff f(y) - f(x) \geq Df_x(y-x)$.

c) Conclude that if a convex function f has a critical point at $x_0 \in U$, then it has a *global* minimum at x_0.

Solution.

a) The 1D Reduction

Let $f_{x,y}$ be convex. We want to show f is convex.

$$\begin{aligned} f_{x,y}((1-t)s_0 + ts_1) &\leq (1-t)f_{x,y}(s_0) + tf_{x,y}(s_1) \\ f(x + [(1-t)s_0 + ts_1](y-x)) &\leq (1-t)f(x + s_0(y-x)) + tf(x + s_1(y-x)) \end{aligned} \tag{9}$$

If we simply set $s_0 = 0$ and $s_1 = 1$, the right side simplifies to $(1-t)f(x) + tf(y)$, and the left side simplifies exactly to $f((1-t)x + ty)$. This immediately proves f is convex! Vice versa, if f is convex, the convexity of $f_{x,y}$ follows by applying the convexity of f to the two points $u = x + s_0(y-x)$ and $v = x + s_1(y-x)$.

b) Tangent Plane Inequality

Recall from Analysis 1 that a 1D function $g \in C^1(I)$ is convex if and only if its derivative g' is monotonically increasing. By part (a), f is convex $\iff$ $f_{x,y}$ is convex $\iff$ $f'_{x,y}$ is increasing $\forall x, y \in U$.

Compute the derivative of the slice using the chain rule for $t \in [0,1]$:

$$f'_{x,y}(t) = Df_{x+t(y-x)}(y-x)$$

By the 1D Mean Value Theorem, there must exist some $s \in (0,1)$ such that:

$$f(y) - f(x) = f_{x,y}(1) - f_{x,y}(0) = f'_{x,y}(s) = Df_{x+s(y-x)}(y-x)$$

Now we apply the increasing property. If f is convex, then $f'_{x,y}(s) \geq f'_{x,y}(0)$.

$$f(y) - f(x) = f'_{x,y}(s) \geq f'_{x,y}(0) = Df_x(y-x)$$

(The proof for the reverse direction involves evaluating the inequality for intermediate points u and v to show that $f'_{x,y}$ must be increasing. See the handwritten notes for the algebraic manipulation.)

c) Critical Points of Convex Functions

Since f is convex, we can always apply the inequality we just proved. If x_0 is a critical point, its differential is zero ($Df_{x_0} = 0$). Plugging this into the inequality for any arbitrary point $y \in U$:

$$f(y) - f(x_0) \geq Df_{x_0}(y - x_0) = 0 \implies f(y) \geq f(x_0)$$

Since this holds for *all* $y \in U$, x_0 is an absolute global minimum! $\square$

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7. The Inverse Function Theorem (IFT)

Theorem (Inverse Function Theorem)**.** Let $U \subseteq \mathbb{R}^n$ be open and $f \in C^k(U, \mathbb{R}^n)$. Let $x_0 \in U$ be a point such that the differential Df_{x_0} is strictly invertible (i.e., $\det(Jf(x_0)) \neq 0$).

Then there exists an open neighborhood $U_0 \subseteq U$ around x_0 such that:

a) $f|_{U_0} : U_0 \to f(U_0)$ is a C^k-diffeomorphism. (It is locally invertible and smooth!)

b) The differential of the inverse is the inverse of the differential:

$$D(f|_{U_0}^{-1})_{f(x)} = (D(f|_{U_0})_x)^{-1} \quad \forall x \in U_0$$

[Gemini-Notiz: Intuition: If the derivative matrix is invertible, it means the function locally acts like a linear transformation that scales or rotates space, but it never *„flattens“* a dimension down to zero. Because no information is destroyed (no dimensions are crushed), you can easily reverse the process—at least locally!]

Question. **a)** If $f : \mathbb{R}^n \to \mathbb{R}^n$ is C^k and a homeomorphism (bijective with continuous inverse), is it necessarily a C^k-diffeomorphism?

b) If $f \in C^\infty(U, \mathbb{R}^n)$ has an everywhere invertible differential (Df_x invertible $\forall x \in U$), is f a C^∞-diffeomorphism onto its image?

Answer. **1. False.** Consider $f : \mathbb{R} \to \mathbb{R}, x \mapsto x^3$. This function is bijective and continuous, and its inverse $f^{-1}(x) = x^{1/3}$ is also continuous. However, f^{-1} is not differentiable at $x = 0$, so it is not a diffeomorphism!

2. False. Local invertibility does not guarantee global invertibility! Consider $f(x) = \cos(x)$ restricted to $\mathbb{R} \setminus \pi\mathbb{Z}$. For any x in this domain, $f'(x) = -\sin(x) \neq 0$, so the differential is everywhere invertible. The IFT guarantees it is locally a diffeomorphism everywhere. But globally, cosine oscillates and hits the same values infinitely many times, so it is obviously not bijective onto $(-1, 1)$!

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8. The Implicit Function Theorem

Theorem (Implicit Function Theorem)**.** **Setup:** Let $n > n - d \geq 1$. Let $f \in C^k(\mathbb{R}^d \times \mathbb{R}^{n-d}, \mathbb{R}^{n-d})$. Suppose $f(x_0, y_0) = 0$ for some point. Suppose $D_y f(x_0, y_0)$, the differential with respect to the y-variables, is bijective (i.e., $J_y f(x_0, y_0)$ is an invertible square matrix).

Consequence: There exist open balls $B_r(x_0) \subseteq \mathbb{R}^d$ and $B_s(y_0) \subseteq \mathbb{R}^{n-d}$, and a C^k function $g : B_r(x_0) \to B_s(y_0)$ such that for all points in this local window:

$$f(x, y) = 0 \iff y = g(x)$$

Furthermore, the Jacobian of this new implicit function is:

$$Jg(x) = -\left(J_y f(x, g(x))\right)^{-1} J_x f(x, g(x))$$

[Gemini-Notiz: The TA wrote: *„The geometrical meaning of this formula eludes me, sorry.“* But it's actually incredibly logical! It's just the Multivariable Chain Rule. We know that locally, $f(x, g(x)) = 0$. If we take the total derivative of both sides with respect to x, the Chain Rule gives: $J_x f + J_y f \cdot Jg = 0$. Now just solve for Jg! Move $J_x f$ to the other side, and multiply from the left by the inverse of $J_y f$. That's it!]

Example (1. The Unit Circle)**.** Consider $F(x, y) = x^2 + y^2 - 1 = 0$. Can we express y as a function of x? Taking the y-derivative: $D_y F(x, y) = 2y$. This is invertible (non-zero) everywhere *except* at $y = 0$ (the points $(-1, 0)$ and $(1, 0)$). Thus, for any point where $y \neq 0$, the IFT guarantees we can locally write $y(x) = \pm\sqrt{1 - x^2}$. At $y = 0$, the tangent is perfectly vertical, so it fails the vertical line test and cannot be written as $y(x)$.

Example (2. The Quintic Equation)**.** Consider the level set $G^{-1}\{0\} = \{(x, y) \in \mathbb{R}^2 : y^5 + y^3 + y + x = 0\}$. Clearly, the origin $(0, 0)$ is a solution since $0^5 + 0^3 + 0 + 0 = 0$. Check the partial derivatives at $(0, 0)$:

$$\begin{aligned} \partial_x G(0, 0) &= 1 \neq 0 \\ \partial_y G(0, 0) &= \left[5y^4 + 3y^2 + 1\right]_{y=0} = 1 \neq 0 \end{aligned} \tag{10}$$

Since $\partial_y G \neq 0$, the IFT guarantees that there exists an open interval around $x = 0$ where we can uniquely write $y = g(x)$.

Why is this theorem so powerful? By the famous Abel-Ruffini Theorem from Galois theory, there is mathematically **no general algebraic formula** to solve a polynomial of degree 5. You literally cannot write down $y(x)$ using standard roots. Yet, the Implicit Function Theorem mathematically guarantees that this function $y(x)$ exists, that it is perfectly smooth, and we can even compute its derivative without ever knowing the function's explicit formula!

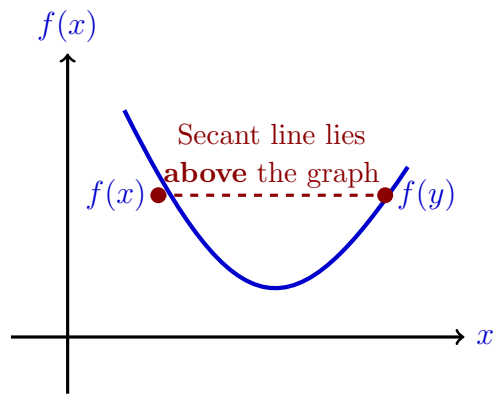
6. Convexity

Definition (Convex Sets and Functions)**.**

- A subset $A \subseteq \mathbb{R}^n$ is **convex** if for any $x, y \in A$ and any $t \in [0, 1]$, the point $p = (1 - t)x + ty$ is also an element of A. (Intuitively: The straight line segment connecting any two points in the set lies entirely within the set.)

- A function $f : A \to \mathbb{R}$ is **convex** if its domain A is a convex set, and for all $x, y \in A$ and $t \in [0, 1]$:

$$f((1-t)x + ty) \leq (1-t)f(x) + tf(y)$$



[Gemini-Notiz: The functional inequality simply states that a secant line between any two points on the graph always stays *„above“* the function itself. Think of a bowl-shaped graph.]

Proposition 5. Let $U \subseteq \mathbb{R}^n$ be open and convex, and $f \in C^2(U)$. The following statements are equivalent:

a) f is a convex function.

b) The Hessian matrix $\mathscr{H}f(x)$ is positive semi-definite for all $x \in U$.

c) $f(y) - f(x) \geq Df_x(y - x)$ for all $x, y \in U$. (The graph always lies above its tangent planes.)

---

7. The Inverse Function Theorem (IFT)

Theorem (Inverse Function Theorem)**.** Let $U \subseteq \mathbb{R}^n$ be open and $f \in C^k(U, \mathbb{R}^n)$. Let $x_0 \in U$ be a point such that the differential Df_{x_0} is invertible ($\det(Jf(x_0)) \neq 0$).

Then there exists an open neighborhood $U_0 \subseteq U$ around x_0 such that:

a) $f|_{U_0} : U_0 \to f(U_0)$ is a C^k-diffeomorphism.

b) The derivative of the inverse is the inverse of the derivative:

$$D(f^{-1})_{f(x)} = (Df_x)^{-1}$$

Question. **a)** If $f : \mathbb{R}^n \to \mathbb{R}^n$ is a C^k homeomorphism, is it necessarily a C^k-diffeomorphism?

b) Does local invertibility everywhere imply global invertibility?

Answer. **1. No.** Example: $f(x) = x^3$. The inverse $x^{1/3}$ is not differentiable at 0.
2. No. Example: $f(x) = (\cos x, \sin x)$ has invertible derivative everywhere but is periodic (not injective).

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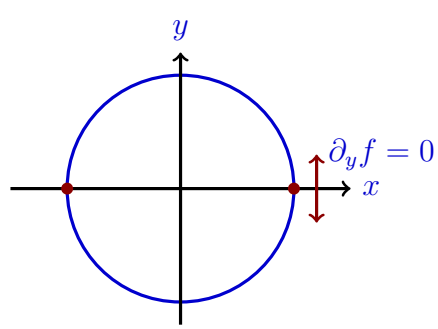
8. The Implicit Function Theorem

Theorem (Implicit Function Theorem)**.** Suppose $f(x_0, y_0) = 0$ and the partial Jacobian $J_y f(x_0, y_0)$ is invertible. Then there exists a unique local function $y = g(x)$ such that $f(x, g(x)) = 0$. The derivative is given by:

$$Jg(x) = -(J_y f)^{-1} J_x f$$

[Gemini-Notiz: This formula is just the Chain Rule in disguise! Differentiating $f(x, g(x)) = 0$ with respect to x gives $J_x f + J_y f \cdot Jg = 0$. Solving for Jg yields the result.]

Example (The Unit Circle)**.** Consider $f(x, y) = x^2 + y^2 - 1 = 0$. We check $\partial_y f = 2y$. This is invertible if $y \neq 0$. Thus, for any point not on the x-axis, we can locally solve for $y = \pm\sqrt{1 - x^2}$.



[Gemini-Warnung: At $y = 0$, the tangent is vertical. The theorem fails here because a small step in x would not uniquely determine y (it could go up or down).]

9. The Implicit Function Theorem

Theorem (Implicit Function Theorem)**.** **Setup:** Let $n > n - d \geq 1$. Let $f \in C^k(\mathbb{R}^d \times \mathbb{R}^{n-d}, \mathbb{R}^{n-d})$. Suppose $f(x_0, y_0) = 0$ for some point $(x_0, y_0) \in \mathbb{R}^d \times \mathbb{R}^{n-d}$[cite: 16]. Suppose the partial differential with respect to the y-variables, $D_y f(x_0, y_0)$, is bijective (i.e., the Jacobian $J_y f(x_0, y_0)$ is an invertible square matrix)[cite: 16].

Consequence: There exist open neighborhoods $U \subseteq \mathbb{R}^d$ of x_0 and $V \subseteq \mathbb{R}^{n-d}$ of y_0, and a unique C^k function $g : U \to V$ such that for all $(x, y) \in U \times V$:

$$f(x, y) = 0 \iff y = g(x)$$

Furthermore, the derivative of this implicit function is given by:

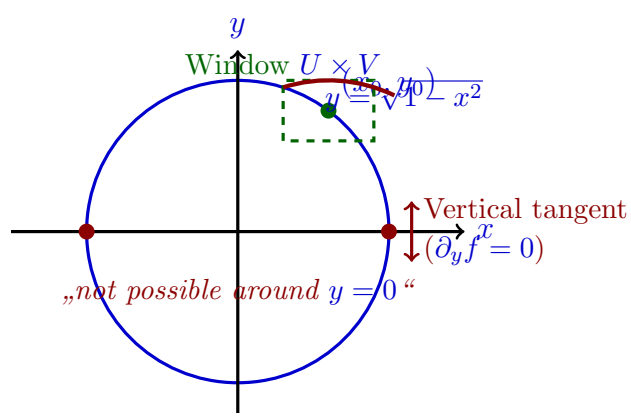
$$Jg(x) = -\left(J_y f(x, g(x))\right)^{-1} J_x f(x, g(x))$$

[Gemini-Notiz: The TA notes mentioned that the geometrical meaning of this derivative formula *„eludes“* them[cite: 16]. But it is actually a beautiful application of the Multivariable Chain Rule! Since we know $f(x, g(x)) = 0$ for all $x \in U$, differentiating both sides with respect to x gives:

$$J_x f + J_y f \cdot Jg = 0$$

Solving for Jg by moving $J_x f$ and multiplying by the inverse of $J_y f$ leads directly to the formula!]

Example (1. The Unit Circle S^1)**.** Consider the level set of $f(x, y) = x^2 + y^2 - 1 = 0$[cite: 16]. We want to know if we can locally solve for y as a function of x. The partial derivative with respect to y is $\partial_y f = 2y$[cite: 16]. This is invertible (non-zero) as long as $y \neq 0$[cite: 16].



[Gemini-Warnung: Around the points $(1, 0)$ and $(-1, 0)$, the tangent to the circle is vertical[cite: 16]. In any small window around these points, a single x-value

would correspond to two different y-values (one on the upper half, one on the lower half). Therefore, we cannot define a unique function $y = g(x)$ there, which matches the fact that $\partial_y f = 0$ at those points.]

Example (2. The Quintic Equation)**.** Consider $G(x, y) = y^5 + y^3 + y + x$[cite: 16]. Does the level set $G^{-1}\{0\}$ define a function $y = g(x)$ around $(0, 0)$? We check the solution: $G(0, 0) = 0^5 + 0^3 + 0 + 0 = 0$. Now check the y-derivative: $\partial_y G = 5y^4 + 3y^2 + 1$[cite: 16]. At $(0, 0)$, we have $\partial_y G(0, 0) = 1 \neq 0$[cite: 16].

By the Implicit Function Theorem, a local function $y = g(x)$ exists! This is a massive result because, according to Abel's Theorem, there is no general formula to solve for the roots of a 5th-degree polynomial. Yet, we know $y(x)$ exists and is smooth.

7. Submanifolds

Recall: A C^k-diffeomorphism is a bijective function whose inverse is also C^k. It essentially acts as a perfectly smooth transformation that stretches and bends space without ever tearing it or creating sharp creases.

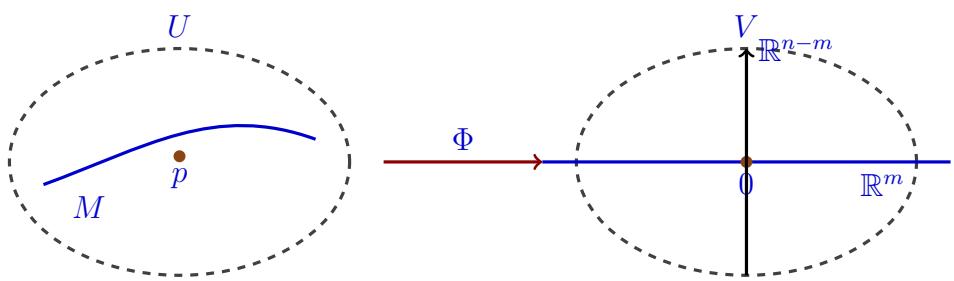
Definition. $M^m \subseteq \mathbb{R}^n$ is a C^k, m-dimensional *submanifold* of $\mathbb{R}^n$ if for any point $p \in M$ there are:

- an *open* neighborhood $U \subseteq \mathbb{R}^n$ of p
- an *open* neighborhood $V \subseteq \mathbb{R}^n$ of 0
- a C^k-diffeomorphism $\Phi : U \to V$ with $\Phi(p) = 0$ (the submanifold chart)

such that

$$\Phi(U \cap M) = (\mathbb{R}^m \times \{0\}) \cap V = \{x \in V : x_{m+1} = \cdots = x_n = 0\}$$

[Gemini-Notiz: The intuition here is beautifully simple: No matter how curved or twisted M looks in the big space $\mathbb{R}^n$, if you zoom in close enough to a point p (inside the window U), you can find a magical map Φ that completely *„irons out“* the wrinkles. It flattens the manifold M so it perfectly aligns with the standard flat coordinate axes of $\mathbb{R}^m$.]



Finding such a chart Φ manually can be extremely difficult. It is often much easier to prove a set is a submanifold by using one of the two standard perspectives: *Implicit Representation* (defining the manifold by what it restricts) or *Local Parametrization* (defining the manifold by building it).

The Implicit Perspective: Cutting down space

To separate the implicit definition from the parametric one, we will use **green for the implicit map F** and **blue for the parametric map f**.

Theorem (Regular value theorem)**.** Suppose $U \subseteq \mathbb{R}^n$ is *open*, $F : U \to \mathbb{R}^l$ is C^k, and

$y \in F(U)$. If the differential

$$DF_x : \mathbb{R}^n \to \mathbb{R}^l$$

is surjective (i.e. $JF(x)$ has full rank) for all $x \in F^{-1}\{y\} = \{\bar{x} \in U : F(\bar{x}) = y\}$, then the level set $F^{-1}\{y\}$ is a C^k submanifold with dimension $n - l$.

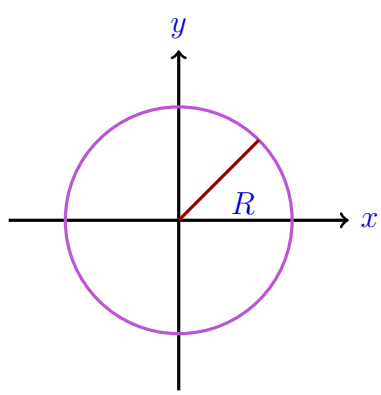
[Gemini-Notiz: Why must it be surjective? Surjectivity of the derivative means the constraints defined by F are truly independent and don't collapse on each other. If the rank dropped, the manifold might have sharp corners, self-intersections, or singular points where it's no longer *„smooth"*.]

Example. Consider $F : \mathbb{R}^2 \to \mathbb{R}$, given by $(x, y) \mapsto x^2 + y^2$.
Then the Jacobian is $JF(x, y) = (2x, 2y)$.
This matrix (a row vector here) has full rank (rank 1) as long as it isn't the zero vector. So it has full rank if $x \neq 0$ or $y \neq 0$, i.e., $\forall (x, y) \in \mathbb{R}^2 \setminus \{0\}$.

For any $R > 0$, the origin $(0, 0) \notin F^{-1}\{R^2\}$. Therefore, the level set

$$F^{-1}\left\{R^2\right\} = \left\{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = R^2\right\}$$

is guaranteed to be a smooth submanifold of dimension $2 - 1 = 1$.



Similarly, $\{x \in \mathbb{R}^n : |x|^2 = R^2\}$ is a submanifold of dimension $n - 1$ for $R > 0$.
This is the classical n-sphere of radius R. **[Gemini-Warnung: Kurzer Terminologie-Check: Eine Sphäre, die im $\mathbb{R}^n$ lebt, ist streng genommen eine $(n-1)$-dimensionale Fläche. Topologen nennen sie daher meist die $(n-1)$-Sphäre, notiert als S^{n-1}. Hier in den Notizen wird sie informell die n-Sphäre genannt (basierend auf der Dimension des Umgebungsraumes).]**

The Connection to Lagrangian Optimization

Recall how we build the Lagrangian $L = f - \sum_{i=1}^{k} \lambda_i g_i$ when optimizing a function $f|_M$ for $f \in C^1(\mathbb{R}^n, \mathbb{R})$. The constraint set M is defined as:

$$g_1, \ldots, g_k : U \to \mathbb{R}, \quad M := \{x \in \mathbb{R}^n : g_1(x) = \cdots = g_k(x) = 0\}$$

For the method of Lagrange multipliers to work, we require that the gradients $(\nabla g_1, \ldots, \nabla g_k)$ are linearly independent $\forall x \in M$.

Let's package all these constraints into a single implicit map $g : \mathbb{R}^n \to \mathbb{R}^k$, where $x \mapsto (g_1(x), \ldots, g_k(x))$. The requirement that the gradients are linearly independent means exactly that the Jacobian:

$$Jg(x) = \begin{pmatrix} \longleftarrow & \nabla g_1(x) & \longrightarrow \\ & \vdots & \\ \longleftarrow & \nabla g_k(x) & \longrightarrow \end{pmatrix}$$

has full rank! Therefore, the constraint surface $M = g^{-1}\{0\}$ is automatically an $(n-k)$-dimensional submanifold by the regular value theorem!

[Gemini-Notiz: Die Notizen erwähnen hier „*ergo* $k \leq n$". Why is this so crucial? Think about what happens if you add constraints. In a 3D room ($n = 3$), one constraint ($k = 1$) restricts you to a 2D floor. Two constraints ($k = 2$) restrict you to a 1D line (where two floors intersect). Three constraints ($k = 3$) restrict you to a 0D point. If you impose $k = 4$ independent constraints in a 3D room, it's physically impossible—the set of solutions is empty! The rank of a matrix can never exceed its number of columns, so for Jg to have full row rank k, we mathematically must have $k \leq n$.]

The Parametric Perspective: Building up space

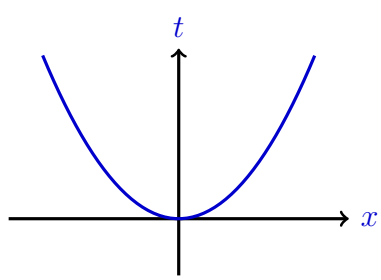
Theorem (Local parametrization)**.** Let $M \subseteq \mathbb{R}^n$. If $\forall p \in M$ there exist an *open* neighborhood $V \subseteq \mathbb{R}^n$ of p, an *open* set $U \subseteq \mathbb{R}^m$ and a C^k map $f : U \to V \cap M$ such that:

a) $Df_x : \mathbb{R}^m \to \mathbb{R}^n$ is injective $\forall x \in U$ (the map doesn't crush dimensions together)

b) f is a homeomorphism (bijective, continuous, and f^{-1} is continuous, ensuring no self-intersections)

then M is a C^k submanifold with dimension m.

Example (Parabola)**.** Let $P = \{(x, t) \in \mathbb{R}^n \times \mathbb{R} : t = |x|^2\}$ (the n-parabola).



We could define this implicitly and use the regular value theorem, but let's build it parametrically. Consider the map:

$$f : \mathbb{R}^n \to \mathbb{R}^n \times \mathbb{R}, \quad x \mapsto (x, |x|^2)$$

We check our conditions:

a) $Jf(x) = \begin{pmatrix} Id_{n\times n} \\ 2x^\top \end{pmatrix}$ clearly has full column rank, so the derivative is injective.

b) f is bijective onto its image and continuous. Its inverse is the canonical projection $\pi : f(U) \to \mathbb{R}^n$, mapping $(x,t) \mapsto x$, which is trivially continuous.

c) $f(\mathbb{R}^n) = P$.

So P is an n-dimensional submanifold.

Note: The graphical representation of a function (like plotting $y = h(x)$) is just a very common, equivalent special case of a local parametrization.

Example (Alternative optimization technique)**.** Let $f : \mathbb{R}^2 \to \mathbb{R}$, given by $(x,y) \mapsto 2x^2 + y^2 - x$.
Let's find the extrema of $f|_{S^1}$ *without* using Lagrange multipliers!

Since S^1 is a 1-dimensional submanifold, we can just parametrize $S^1 = \{(x,y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$ using an angle θ:

$$\begin{pmatrix} x(\theta) \\ y(\theta) \end{pmatrix} = \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}$$

We substitute this parametrization directly into our objective function:

$$\begin{aligned} f(\theta) = f(x(\theta), y(\theta)) &= 2\cos^2\theta + \sin^2\theta - \cos\theta \\ &= 2\cos^2\theta + (1 - \cos^2\theta) - \cos\theta \\ &= \cos^2\theta - \cos\theta + 1 \end{aligned} \tag{11}$$

Now it's just a standard Analysis 1 problem! Setting the derivative to zero:

$$f'(\theta) = -2\cos\theta\sin\theta + \sin\theta = 0$$

Factoring out $\sin\theta$:

$$\sin\theta(1 - 2\cos\theta) = 0$$

This gives two cases where the product is zero:

a) $\sin\theta = 0 \implies \theta_1 \in \{0, \pi\}$

b) $(1 - 2\cos\theta) = 0 \implies \cos\theta = \frac{1}{2} \implies \theta_2 \in \left\{\frac{\pi}{3}, \frac{5\pi}{3}\right\}$

Plugging these angles back into our parametrization $(x,y) = (\cos\theta, \sin\theta)$, the extrema on the circle are located at:

a) $(\cos\theta_1, \sin\theta_1) = (\pm 1, 0)$

b) $(\cos\theta_2, \sin\theta_2) = \left(\frac{1}{2}, \pm\frac{\sqrt{3}}{2}\right)$

8. Tangent and Normal space

Suppose $M^m \subseteq \mathbb{R}^n$ is a submanifold and $f : U \to f(U) \subseteq M$ is a local parametrization around some point $p \in f(U)$ with $f(q) = p$.

The $n \times m$-matrix of partial derivatives

$$Jf(q) = \begin{pmatrix} | & & | \\ \partial_1 f(q) & \dots & \partial_m f(q) \\ | & & | \end{pmatrix}$$

has full rank, meaning the column vectors $\partial_i f(q)$ are linearly independent.

[Gemini-Notiz: Think of the vectors $\partial_i f(q)$ as the *„velocity vectors“* you get if you walk along the grid lines of your parameter space U. Since they are linearly independent, they span a perfectly flat m-dimensional plane that grazes the manifold at p.]

Definition (Tangent space)**.** For M, f as above, the Tangent space of M at $p \in M$ is the space spanned by these velocity vectors:

$$T_p M = \operatorname{span}\{\partial_1 f(q), \dots, \partial_m f(q)\} = Df_q(\mathbb{R}^m)$$

Remarkably, this space is completely independent of the specific parametrization f you chose to use!

Example. We parametrize the upper hemisphere

$$S^2_+ = \left\{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = 1, z > 0\right\}$$

graphically as a function of x and y:

$$f(x, y) = \begin{pmatrix} x \\ y \\ \sqrt{1 - x^2 - y^2} \end{pmatrix}$$

Let's determine the tangent space $T_{e_3} S^2_+$ at the *„North Pole“*, where $e_3 = (0, 0, 1)$. This corresponds to the parameters $(x, y) = (0, 0)$.

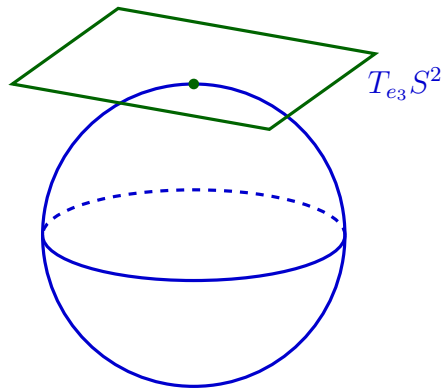
First, we calculate the partial derivatives and evaluate them at the origin:

$$\partial_x f\Big|_{(x,y)=(0,0)} = \begin{pmatrix} 1 \\ 0 \\ \frac{-x}{\sqrt{1-x^2-y^2}} \end{pmatrix}\Bigg|_{(x,y)=(0,0)} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} =: e_1$$

$$\partial_y f\Big|_{(x,y)=(0,0)} = \begin{pmatrix} 0 \\ 1 \\ \frac{-y}{\sqrt{1-x^2-y^2}} \end{pmatrix}\Bigg|_{(x,y)=(0,0)} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} =: e_2$$

So, the tangent space at the North Pole is perfectly flat against the xy-plane:

$$T_{e_3}S^2_+ = T_{e_3}S^2 = \text{span}\{e_1, e_2\} = \mathbb{R}^2 \times \{0\}$$



The Grand Synthesis: Tangent vs. Normal

Now, let's look at the beautiful relationship between our two perspectives. Suppose $F : U^n \to \mathbb{R}^{n-m}$ is an *implicit representation* of a submanifold $M^m = F^{-1}\{0\}$, and $f : V^m \to U$ is a *local parametrization* of that same manifold.

Then, by definition of these two constructs, three things are true:

a) $JF(x) = \begin{pmatrix} \longleftarrow & \nabla F_1(x) & \longrightarrow \\ & \vdots & \\ \longleftarrow & \nabla F_{n-m}(x) & \longrightarrow \end{pmatrix}$ has full rank. Thus, the gradients $\{\nabla F_i(x)\}_{i=1,\dots,n-m}$ are linearly independent.

b) $T_{f(q)}M = \text{span}\{\partial_i f(q)\}$ as we just discussed.

c) $F(f(x)) = 0 \quad \forall x \in V$. This is the crucial link: because the image of the parametrization f lies entirely on the manifold, feeding $f(x)$ into the constraint equation F must always output exactly zero!

From point **c)**, we can apply the multivariable chain rule to see what happens to the derivatives:

$$J(F \circ f)(q) = JF(f(q)) \cdot Jf(q) = \begin{pmatrix} \longleftarrow & \nabla F_1(f(q)) & \longrightarrow \\ & \vdots & \\ \longleftarrow & \nabla F_{n-m}(f(q)) & \longrightarrow \end{pmatrix} \begin{pmatrix} | & & | \\ \partial_1 f(q) & \dots & \partial_m f(q) \\ | & & | \end{pmatrix} = 0$$

Now, look closely at what this matrix multiplication is actually doing (!!!):

$$(JF(f(q)) \cdot Jf(q))_{ij} = \langle \nabla F_i(f(q)), \partial_j f(q) \rangle = 0 \quad \forall i = 1, \dots, n-m, \quad j = 1, \dots, m$$

Every single gradient vector of F has a dot product of zero with every single tangent vector from f. So the vectors $\nabla F_i(f(q))$ are all strictly perpendicular (normal) to the tangent space $T_{f(q)}M$.

Since by point **a)**, we have exactly $n-m$ linearly independent vectors, and they are all perpendicular to our m-dimensional tangent plane, they perfectly span the remaining dimensions of $\mathbb{R}^n$. This leads us to define:

Definition (Normal space)**.** Let $M^m \subseteq \mathbb{R}^n$ be a submanifold and $F : U^n \to \mathbb{R}^{n-m}$ an implicit representation where $M = F^{-1}\{0\}$. Then the normal space of M at $p \in M$ is spanned by the gradients of the constraints:

$$N_pM := \operatorname{span}\{\nabla F_1(p), \ldots, \nabla F_{n-m}(p)\}$$

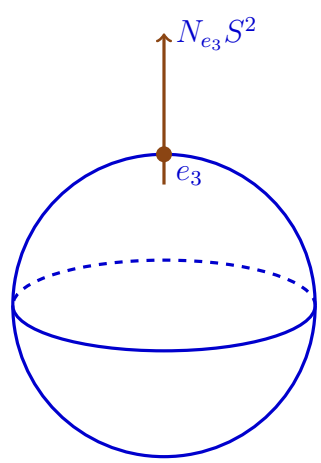
Example. Let's represent the sphere S^2 implicitly:

$$S^2 = F^{-1}\{1\} = \left\{(x,y,z) \in \mathbb{R}^3 : F(x,y,z) = 1\right\}$$

where the constraint is $F(x,y,z) = x^2 + y^2 + z^2$.

We determine the normal space $N_{e_3}S^2$ at the North Pole $e_3 = (0,0,1)$:

$$\nabla F(0,0,1) = (2x, 2y, 2z)\Big|_{(x,y,z)=(0,0,1)} = (0,0,2) = 2e_3$$

So, the normal space points straight up along the z-axis:

$$N_{e_3}S^2 = \operatorname{span}\{e_3\} = \{0\} \times \{0\} \times \mathbb{R}$$